

Narrative Team at SemEval-2026 Task 5: Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

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Abstract

This paper describes our system for SemEval-2026 Task 5, which focuses on predicting the plausibility of narrative continuations given contextual and semantic information. The task requires assigning a real-valued plausibility score to short narrative texts based on aggregated human judgments. Our approach explores two modeling paradigms: (i) a pretrained transformer-based regression model using DistilBERT fine-tuned on the task data, and (ii) a lightweight neural baseline based on a bidirectional LSTM trained from scratch or initialized with GloVe embeddings. Input representations combine a disambiguated meaning definition with narrative context and target sentence, separated by a special token. Experimental results show that the transformer model achieves the strongest overall performance, reaching an accuracy-within-standard-deviation of 0.54 on the official test set, while the best LSTM configuration reaches 0.52. Although transformer-based models remain dominant, our findings suggest that simpler architectures can remain competitive when carefully tuned. We provide an analysis of model variants, error patterns, and trade-offs between performance and computational cost, highlighting challenges related to fine-grained plausibility estimation in narrative understanding tasks.

1 Introduction

Narrative understanding requires more than identifying a single correct interpretation: human readers often tolerate ambiguity and assign graded plausibility to alternative word senses depending on context, underspecification, and personal expectations. SemEval-2026 Task 5 targets this phenomenon by framing plausibility as a scalar prediction problem: systems must estimate how plausible a given sense of an ambiguous homonym is within a short story context, based on human ratings.

Most competitive approaches to graded semantic judgments in NLP rely on pretrained transformer encoders fine-tuned for regression or ranking. Distilled transformer variants offer a favorable trade-off between compute and performance, often retaining much of the representational power of larger encoders. In addition to a transformer solution, we also explore a simpler recurrent baseline based on a bidirectional LSTM to better understand how much of the task can be captured by sequence modeling without large-scale pretraining.

Our main contributions are: (1) a straightforward regression formulation for plausibility prediction using DistilBERT, aligned with the task’s ordinal ratings; (2) a direct comparison between a transformer regressor (DistilBERT) and a BiLSTM regressor under the shared-task metrics; and (3) an ablation-style analysis across recurrent variants highlighting trade-offs between correlation and tolerance-based accuracy.

2 Background

2.1 Task Description

SemEval-2026 Task 5 introduces AmbiStory, a dataset of five-sentence short stories designed to probe narrative-based word sense plausibility. Each story setup contains: (1) a precontext of three sentences grounding the scenario; (2) an ambiguous sentence (fourth sentence) containing a target homonym with two widely different plausible senses; and (3) an optional ending (fifth sentence) that may bias the interpretation toward one sense.

For each story and each candidate sense, human annotators rate plausibility on a 1–5 scale. The task requires predicting the human-perceived plausibility of a given sense in context. Each story setup appears with two candidate senses and three context variants (ending A, ending B, or no ending),

resulting in six annotated samples per setup, with at least five human ratings per sample.

2.2 Evaluation Metrics

The shared task uses two primary evaluation measures:

- **Spearman correlation (ρ):** measures rank correlation between predicted scores and average human ratings.
- **Accuracy within standard deviation (Acc@SD):** the proportion of predictions whose absolute error is within the (minimum-clipped) standard deviation of annotator ratings for that sample, reflecting agreement sensitivity.

These metrics reward complementary behaviors: correlation prioritizes correct relative ordering, while Acc@SD rewards making predictions consistent with annotator consensus and tolerates higher variance where humans disagree.

2.3 Related Work

Most Word Sense Disambiguation (WSD) work formulates disambiguation as selecting a single correct sense. However, human interpretations are often graded, motivating approaches that predict plausibility or applicability scores over candidate senses. Erk and McCarthy (2009) study graded sense assignment using human ratings and show that modeling sense plausibility provides a more faithful representation of ambiguity than single-label prediction.

Recent supervised WSD systems largely rely on pretrained Transformers. GlossBERT leverages contextualized encoders and sense gloss information by framing WSD as context–gloss matching, demonstrating strong performance and highlighting the importance of target-aware contextual representations. While our task does not require gloss matching, these findings support using Transformer encoders for fine-grained semantic plausibility modeling.

SemEval-2026 Task 5 extends graded sense prediction to narrative contexts, requiring systems to estimate plausibility ratings (1–5) for candidate senses in short stories. The organizers report strong LLM baselines that remain below human performance, indicating the task’s difficulty. We therefore compare a lightweight Transformer regressor with BiLSTM baselines to study the trade-off between contextual capacity and model simplicity.

3 System Overview

Our system architecture is shown in Figure 1. Both models consume the same input text representation and output a single plausibility score.

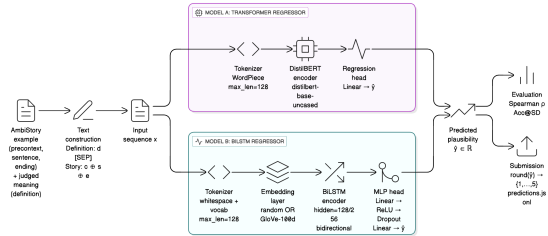


Figure 1: Architecture for plausability regression.

3.1 Model Architecture

Each labeled example provides (a) a disambiguated sense definition and (b) a narrative context comprising precontext, ambiguous sentence, and optionally an ending. We convert each example to a single sequence:

Definition: judged_meaning [SEP] Story: precontext + sentence + ending

3.2 Transformer regressor: DistilBERT

Our primary submission fine-tunes DistilBERT for regression by using a sequence classification head with num_labels = 1. DistilBERT is a compressed transformer trained via knowledge distillation from BERT, offering a smaller, faster encoder while retaining strong performance.

We use distilbert-base-uncased as the encoder with WordPiece tokenization and truncation/padding to 128 tokens, producing a single scalar plausibility prediction from the regression head.

For the BiLSTM baseline, we experiment with two embedding initializations: (i) randomly initialized trainable embeddings learned from scratch, and (ii) embeddings initialized with pretrained GloVe vectors and fine-tuned during training.

3.3 Recurrent baseline: BiLSTM regressor

To compare against large pretrained encoders, we implement a word-level BiLSTM model in PyTorch. The model maps tokens to embeddings, applies a bidirectional LSTM, and uses a small feed-forward head to predict a scalar score.

Our BiLSTM architecture consists of an embedding layer, either randomly initialized and trained from scratch or initialized with pretrained GloVe

vectors, a bidirectional LSTM encoder, and a lightweight prediction head with a dense ReLU layer and dropout followed by a final linear projection to a single scalar plausibility score.

The BiLSTM produces the final forward and backward hidden states, which are concatenated into h . We apply an MLP with ReLU and dropout and map to a scalar plausibility score $\hat{y}$.

$$h = [h_T^{\rightarrow}; h_1^{\leftarrow}] \quad (1)$$

We apply an MLP with ReLU and dropout and map to a scalar plausibility score $\hat{y}$:

$$\hat{y} = W_2 \text{Dropout}(\text{ReLU}(W_1 h + b_1)) + b_2. \quad (2)$$

4 Experimental Setup

We follow the task’s provided training and validation sets and additionally create an internal split for model selection. Specifically, we concatenate the provided train and development sets into a single pool and perform an 85/15 train-validation split with a fixed random seed for reproducibility.

Each training instance includes:

- **label** (float): the average plausibility rating;
- **stdev** (float): the standard deviation across annotators, used for Acc@SD computation.

4.1 Training Details

We fine-tune DistilBERT using Hugging Face Transformers. The hyperparameters are listed in Table 3.

Parameter	Value
Epochs	3
Batch Size	16
Learning Rate	2×10^{-5}
Weight Decay	0.01
Max Sequence Length	128

Table 1: DistilBERT hyperparameters

During evaluation, we compute Spearman (ρ) between predicted scores and gold averages, and Acc@SD using the stored per-sample standard deviation.

Table 2 shows the hyperparameters of BiLSTM training. We use a simple word tokenizer (lowercasing, removing the literal “[sep]”, whitespace split). We build a vocabulary from the training corpus and keep tokens with frequency > 1 ; remaining tokens

Parameter	Value
Loss	MSE
Optimizer	Adam
Learning Rate	1×10^{-3}
Epochs	50
Dropout	0.3

Table 2: BiLSTM hyperparameters

map to $\langle \text{UNK} \rangle$. Inputs are padded/truncated to 128 tokens.

For GloVe variants, we initialize embedding rows with pretrained vectors where available and sample random vectors for missing tokens.

We round model outputs to the nearest integer rating for submission.

5 Results

Table 3 reports our evaluation-server results.

System	Acc@SD
DistilBERT regressor	0.54
BiLSTM (no-GloVe)	0.52
BiLSTM (GloVe-initialized)	0.48

Table 3: Testing Phase results

On our internal 85/15 validation split, the BiLSTM family showed steady gains in Spearman correlation as training progressed, peaking around the mid-to-late epochs depending on configuration. Increasing hidden size from 128 to 256 generally improved stability, while adding extra dense layers did not reliably improve results and sometimes reduced Acc@SD, suggesting mild overfitting.

Interestingly, GloVe initialization substantially reduced training loss but did not translate into better correlation or Acc@SD, and underperformed the random-embedding baseline on the official evaluation. A likely explanation is domain mismatch: plausibility judgments in AmbiStory depend heavily on narrative coherence and fine-grained contextual cues, which may not be captured by static embeddings alone.

6 Conclusion

We present systems for SemEval-2026 Task 5, comparing a DistilBERT regressor against BiLSTM baselines with architectural and embedding variants. DistilBERT achieved the best official testing-phase Acc@SD (0.54), while our strongest BiLSTM variant reached 0.52, showing that simpler

recurrent models can be competitive with careful tuning.

Future work can include pairwise modeling across senses, improved handling of ending-conditioned shifts in plausibility, and leveraging stronger pretrained encoders or instruction-tuned models for definition–context reasoning.

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